Date: ___/___/___

Linear Regression

- Analyze specific relationships b/w variables
- Done to gain information.

Primarily used to:

- Estimate relationship that exists, on avg, b/w dep and ind. variable. (output)
- Predict val of dep variable for given val of ind. variable. (inputs)

Assumptions:

- Linear functional form
- fixed ind. variables
- Representative ~~model~~ sample (no omitted vars)
- Equality of variance of errors.
- No multicollinearity
- No autocorrelation of errors
- No outlier distortion.

Date: ___|___|___

Model : $y = \underbrace{b_0}_{\text{y-int}} + \underbrace{b_1 x}_{\text{slope}} + \underbrace{\epsilon}_{\text{error}}$

$y = a + mx$

Least Squares Estimation.

$$\beta_1 = \frac{\Sigma (x_i - x')(y_i - y')}{\Sigma (x_i - x')^2}$$

$\beta_0 = \bar{y} Y' - \beta_1 x'$

Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
import pandas as pd.

model = LinearRegression().fit(x, y)

r2 = model.score(x, y)

y_pred = model.predict(x)
```

CamScanner